

Discrete Goalpost Widening

How Pity Cap Manipulation Extends the Uncertainty Window
to Maximise Spending Pressure in Gacha Loot Systems

Abstract

This paper presents a Monte Carlo analysis of the Sanctum loot system, synthesising two independent findings to expose the monetisation function of the S1 pity cap. A Shapley-value variance decomposition reveals that S1 timing dominates outcome uncertainty in the critical early-to-mid session (accounting for up to 68.9% of explained variance at roll 50). A parameter sweep across pity caps from 10 to 80 rolls shows that each 10-roll increment extends the zone of peak uncertainty by 11–14 rolls. The synthesis demonstrates that the current 80-roll cap places the *entire* uncertainty window over both player archetypes: casual players who exit at the 150-point redemption threshold (roll ~ 38) and skin-seekers who commit to the pity cap for the max prize (roll 80). For casual players, only 16.5% trigger the pity guarantee by their exit point; the cap functions as a psychological anchor rather than a practical mechanic. For skin-seekers, the cap sets the price of the guaranteed outcome at £214 rather than the £85–£108 it would cost at caps of 30–40. The cumulative effect of widening the cap from 10 to 80 rolls generates 79 additional anxious rolls, representing £213 in potential additional spend per player.

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1 Introduction and Motivation

Gacha loot systems are a dominant monetisation strategy in free-to-play games, generating revenue through randomised item acquisition. A core design element in many such systems is the *pity mechanic*: a guaranteed high-value outcome after a specified number of unsuccessful rolls. The pity cap is typically presented to players as a consumer-friendly safeguard — a ceiling on bad luck. This paper examines whether the cap also serves a second, less visible function: as a precision tool for calibrating spending pressure.

The Sanctum loot system awards items from a pool of 753 possibilities distributed across three subsets:

- **Subset 1 (S1):** 1 one-time item + 1 permanent item, total selection probability 0.50%. The one-time S1 item is the highest-value outcome, and its acquisition triggers a *transform* that permanently alters the probability distribution. Pity guarantees this transform within N rolls (default $N = 80$).
- **Subset 2 (S2):** 10 one-time items + 1 permanent item, total probability 10.00%.
- **Subset 3 (S3):** 742 items (5 permanent + 263 emotes + 474 icons), total probability 89.50%.

Each roll selects one item according to a probability distribution that evolves dynamically: one-time items are removed after collection, and their probability mass is redistributed among remaining items within the same subset. The mathematical foundations of this process — two-stage sampling, inverse CDF sampling, and probability redistribution — are detailed in the companion document `simulation_foundations.pdf`.

The central question of this paper is: *how does varying the S1 pity cap affect both the statistical properties of player outcomes and the psychological conditions under which spending decisions are made?*

This is addressed by connecting two independent analyses:

1. A **Shapley-value variance decomposition** that quantifies how much of the total outcome variance is attributable to S1 timing, S2 timing, and S3 item selection at each roll.
2. A **pity cap parameter sweep** that measures expected point accumulation, variance, and rolls-to-target across caps from 10 to 80 in increments of 10.

The synthesis of these two findings reveals the mechanism termed *discrete goalpost widening*: incremental cap increases that each appear modest in isolation but cumulatively extend the zone of maximum outcome uncertainty across the player’s entire natural session.

2 Method

All simulations use 300,000 independent trials of up to 1,000 rolls each, executed on an NVIDIA GeForce RTX 5060 Ti via CuPy for GPU-accelerated computation. The random seed is fixed at 42 for reproducibility. Full implementation details are in `expected_points.py`.

2.1 Variance Decomposition

The total variance of cumulative points at each roll is decomposed into contributions from three sources of randomness:

- **S1 timing:** when the S1 transform occurs.
- **S2 timing:** when all S2 one-time items are collected.
- **S3 selection:** which specific S3 items are drawn on each roll.

Following a 2^3 factorial design, eight simulation configurations are run: a baseline with all sources free, three single-fix variants (each fixing one source at its baseline median), three double-fix variants, and a fully-fixed variant. The contribution of each source is computed via Shapley values, which account for interaction effects by averaging marginal contributions across all possible orderings.

Formally, for source i , the Shapley value is:

$$\phi_i = \sum_{S \subseteq \{1,2,3\} \setminus \{i\}} \frac{|S|! (2 - |S|)!}{3!} [V(S \cup \{i\}) - V(S)] \quad (1)$$

where $V(S)$ denotes the variance reduction when the sources in S are fixed. This ensures the decomposition is unique, symmetric, and sums to the total explained variance.

A sensitivity analysis confirms that the decomposition is robust to the choice of fix-point: results are stable across the 25th, 50th, and 75th percentile trigger rolls for both S1 and S2.

2.2 Pity Cap Parameter Sweep

The S1 pity cap is swept across $\{10, 20, 30, 40, 50, 60, 70, 80\}$ rolls, running 300,000 simulations at each value. For each configuration, the following are recorded:

- Mean and median cumulative points at checkpoint rolls (10, 20, 30, 50, 80, 100, 250).
- Standard deviation per roll.
- The fraction of simulations that have triggered S1 by each roll.
- Median rolls required to reach point targets (50, 100, 150, 200, 300, 500 points).

2.3 Synthesis: The Uncertainty Window

The synthesis phase connects both analyses by measuring S1’s variance contribution *at each pity cap*. For each cap value, a baseline simulation and a fix-S1 variant (forcing S1 to trigger at its median roll for that cap) are run. S1’s variance share at roll r is:

$$\text{S1 share}(r) = \frac{V_{\text{baseline}}(r) - V_{\text{fix-S1}}(r)}{V_{\text{baseline}}(r)} \quad (2)$$

The **uncertainty window** is defined as the contiguous roll range where S1’s smoothed variance share exceeds 30% (using a 10-roll moving average). This threshold captures the zone where a single binary event — whether S1 has triggered — accounts for nearly a third or more of all outcome variation.

3 Results

3.1 S1 Timing Dominates Early-to-Mid Session Variance

The Shapley decomposition reveals a clear pattern: S1 timing is the dominant source of outcome uncertainty throughout the critical early-to-mid game, despite S1 items comprising only 0.5% of the probability mass.

Table 1: Shapley variance decomposition at the default pity cap (80 rolls). V_{total} is baseline variance; percentages are shares of explained variance ($V_{\text{total}} - V_{\text{residual}}$).

Roll	V_{total}	V_{residual}	S1	S2	S3	Sum
10	253	27	28.4%	0.0%	71.6%	100.0%
25	952	46	51.2%	0.6%	48.2%	100.0%
50	2,903	97	68.9%	0.4%	30.7%	100.0%
100	20,951	11,911	55.6%	25.9%	18.5%	100.0%
250	44,774	32,624	47.7%	17.8%	34.5%	100.0%

At roll 25 — roughly where many players are actively engaged — S1 timing accounts for 51.2% of explained variance. By roll 50 this rises to 68.9%. The implication is stark: the single most important factor determining whether a player’s session feels “lucky” or “unlucky” is whether S1 has triggered.

Critically, the interaction effects from the 2^3 factorial analysis confirm that S1 timing acts nearly independently of S2 and S3. All two-way and three-way interaction terms are below 1.5% of baseline variance at every checkpoint. This independence means the pity cap is a *clean, isolated lever*: adjusting it modifies spending pressure without creating unpredictable knock-on effects elsewhere in the system.

3.2 Pity Cap Sweep: Point Accumulation and Cost

Table 2 summarises mean cumulative points at key checkpoints across all pity cap values.

Table 2: Mean cumulative points at selected checkpoints by S1 pity cap. All values from 300,000 simulations.

Cap	Roll 10	Roll 20	Roll 30	Roll 50	Roll 80	Roll 100
10	39.7	346.5	653.8	1,270.3	2,249.9	2,965.5
20	42.3	81.3	146.8	508.4	1,197.8	1,651.7
30	42.3	84.6	123.9	254.7	734.8	1,166.6
40	42.3	84.6	127.4	232.7	673.8	924.1
50	42.3	84.6	127.4	212.3	472.7	866.8
60	42.3	84.6	127.4	215.8	452.5	683.4
70	42.3	84.6	127.4	215.8	433.5	663.8
80	42.3	84.6	127.4	215.8	414.8	645.2

Players fall into two broad categories. *Skin-seekers* enter the system aiming for the S1 max prize (the Viego skin), which requires reaching the pity cap at roll 80 — a cost of £213.98 using an optimal package combination. *Casual players* engage without committing to the full pity distance and typically accumulate around 150 expected points (roll ~ 38 , £102.49) before stopping. The gap between these two endpoints — £111.49 — defines the *sunk-cost trap* for casual players who did not initially intend to reach pity but find themselves anchored by prior spend.

Table 3 shows median rolls to reach point milestones, with GBP equivalents.

Table 3: Median rolls to reach point targets, with DP-optimal purchase cost.

Cap	50 pts	100 pts	150 pts	200 pts	300 pts	500 pts
10	14 (£40)	20 (£58)	20 (£58)	20 (£58)	20 (£58)	30 (£85)
20	13 (£39)	26 (£74)	38 (£102)	40 (£108)	40 (£108)	56 (£152)
30	13 (£39)	25 (£71)	38 (£102)	49 (£133)	60 (£164)	60 (£164)
40	13 (£39)	25 (£71)	38 (£102)	50 (£134)	67 (£178)	80 (£214)
80	13 (£39)	25 (£71)	38 (£102)	50 (£134)	68 (£182)	91 (£244)

A key observation: at the 150-point milestone, median rolls are *identical* across caps 20–80. For casual players, the cap does not materially change how quickly they reach this point. For skin-seekers, however, the cap *is* the target: the pity guarantee at roll 80 is the entire reason they are spending. The cap determines both the cost of entry and the length of the uncertainty they must endure to reach it.

3.3 The Uncertainty Window

Combining the variance decomposition with the pity cap sweep produces the central result. Table 4 shows the uncertainty window — the roll range where S1 timing drives more than 30% of total variance — at each cap.

Table 4: Uncertainty window by pity cap. Width is the number of rolls where S1 timing accounts for >30% of total outcome variance.

Cap	Median S1	Start	End	Width	Peak share
10	10	—	—	0	14.6%
20	20	18	23	6	35.1%
30	30	18	37	20	49.6%
40	40	18	49	32	58.3%
50	50	18	60	43	64.3%
60	60	18	72	55	66.2%
70	70	18	83	66	66.1%
80	80	18	96	79	66.1%

At cap 80, the uncertainty window spans rolls 18–96. For casual players (rolls 1–38), the entire session falls within this zone. For skin-seekers (rolls 1–80), they spend their entire journey inside the region of maximum outcome uncertainty before the pity guarantee resolves it at the last possible moment.

Each 10-roll cap increment extends the window by 11–14 rolls:

Table 5: Marginal impact of each 10-roll cap increase on the uncertainty window, with DP-optimal purchase cost for those extra rolls.

Change	Old width	New width	Δ rolls	Extra spend (£)
10 $\rightarrow$ 20	0	6	+6	19.25
20 $\rightarrow$ 30	6	20	+14	39.98
30 $\rightarrow$ 40	20	32	+12	36.00
40 $\rightarrow$ 50	32	43	+11	31.50
50 $\rightarrow$ 60	43	55	+12	36.00
60 $\rightarrow$ 70	55	66	+11	31.50
70 $\rightarrow$ 80	66	79	+13	39.24
Total			+79	212.72

No single increment exceeds 14 rolls. Each change, viewed in isolation, appears modest. The cumulative effect is an expansion of 79 anxious rolls representing £212.72 in potential additional spend per player.

4 Synthesis: The Discrete Goalpost Widening Effect

The results converge on a single mechanism, described here in three parts.

4.1 The Monetisation Geometry

At 400 RP per roll, using optimal package combinations, two distinct spending trajectories emerge:

- **Skin-seekers** target the S1 pity guarantee and budget for roll 80: £213.98. This is their *intended* spend from the outset.
- **Casual players** accumulate ~ 150 points by roll ~ 38 : £102.49. At this threshold the accumulated points can be redeemed for a specific reward outside the loot system, providing a natural exit point.
- The gap between these endpoints: £111.49.

For **casual players**, the gap is the monetisation sweet spot: large enough to extract significant additional spend, but framed as “finishing what you started” rather than initiating a new purchase. At roll 38 with cap 80, only 16.5% of players have triggered S1. Those who stop here leave without the max prize, and the pity cap functions as a *psychological anchor* — a visible guarantee that reframes quitting as a loss.

For **skin-seekers**, the monetisation operates differently. These players enter the system already committed to reaching the pity cap. The uncertainty window does not need to *trick* them into spending more — they have already accepted the full cost. Instead, the cap’s function is to *set the price*: an 80-roll cap means £214 rather than the £85–£108 it would cost at caps of 30–40. The discrete goalpost widening effect determines *how much* the skin costs, not whether the player pursues it. A player who would have paid £85 for the skin at cap 30 must now pay £214 for the identical outcome at cap 80.

4.2 Package Granularity and Forced Overspend

The spending pressure created by the pity cap does not operate in isolation — it is compounded by the structure of the RP package store. Table 6 lists the available packages.

Table 6: RP packages with effective rolls per package. Only the 2,800 RP package yields a whole number of rolls (400 RP each).

RP	GBP	Rolls covered	GBP/roll
575	£4.50	1.44	£3.13
1,380	£10.25	3.45	£2.97
2,800	£19.99	7.00	£2.86
4,500	£31.50	11.25	£2.80
6,500	£44.99	16.25	£2.77
13,500	£88.99	33.75	£2.64
33,500	£220.00	83.75	£2.63
60,200	£385.00	150.50	£2.56

Two properties of this structure deserve attention.

1. Fractional Roll Counts

Most packages yield a fractional number of rolls. Only the 2,800 RP package divides cleanly (7.00 rolls exactly). All others produce a fractional remainder, meaning that *no single purchase of those packages* can be exactly exhausted through rolling — there is always a residual RP balance.

Even with an optimal combination chosen by the DP algorithm, achieving a specific roll target still requires overspending slightly. Consider both player types:

Casual player targeting 150 points (~ 38 rolls = 15,200 RP):

- Optimal combination: $1 \times 13,500$ RP (£88.99) + 3×575 RP (£13.50) = 15,225 RP — 25 RP surplus — total £102.49.
- Alternative: $1 \times 13,500 + 1 \times 1,380 + 1 \times 575$ RP (£103.74): yields 15,455 RP — 255 RP surplus, £1.25 more expensive.

Skin-seeker targeting the pity cap (80 rolls = 32,000 RP):

- Optimal combination: $2 \times 13,500$ RP (£177.98) + $1 \times 4,500$ RP (£31.50) + 1×575 RP (£4.50) = 32,075 RP — 75 RP surplus — total £213.98.
- Alternative: $1 \times 33,500$ RP (£220.00): yields 33,500 RP — 1,500 RP (3.75 rolls) surplus, £6.02 more expensive.

Even the optimal purchase produces residual RP — a *re-engagement hook* that feels wasteful to abandon. The casual player’s 25 RP surplus and the skin-seeker’s 75 RP surplus are each insufficient for even one additional roll (which costs 400 RP), yet both create a balance that nudges toward a further top-up purchase.

2. Non-Uniform Increments

The package sizes (575, 1,380, 2,800, 4,500, 6,500, 13,500, 33,500, 60,200) are not multiples of a common base. If packages were sold in clean multiples of 400 RP, a player could calculate exactly how many rolls they were buying and purchase accordingly. Instead, the irregular increments introduce a *currency obfuscation layer*: the conversion from pounds to RP to rolls requires mental arithmetic that most players will approximate rather than compute precisely.

This approximation systematically favours the house. A player who reasons “the 13,500 RP package should cover 38 rolls” is 200 RP short ($13,500 < 15,200$). A player who guesses “two

6,500 RP packages” has 13,000 RP — still 2,200 RP short of 38 rolls. In both cases, the player either stops short of their target or buys an additional package, generating surplus RP that seeds further spending.

Interaction with the Pity Cap

The package structure and the pity cap form a complementary system. The pity cap determines *how many rolls* a player desires by anchoring a guaranteed outcome at a specific distance. The package structure ensures that *purchasing those rolls* always involves waste. Together, they create a two-layer extraction mechanism:

1. The uncertainty window pressures the player to want more rolls than they originally intended.
2. The package granularity ensures that buying those rolls produces leftover currency that encourages yet another session.

This interaction means that the real cost of engaging with the system always involves some overspend beyond the exact RP required. Even with optimal package selection, the casual player pays £102.49 for 15,225 RP when they need 15,200 RP, and the skin-seeker pays £213.98 for 32,075 RP when they need 32,000 RP. In both cases, residual RP either goes unused or seeds further spending.

4.3 Three Psychological Functions of the Pity Cap

1. Sunk Cost Anchor

For **casual players**, reaching 150 points (roll 38) creates a framing where they are 42 rolls away from a guaranteed outcome. Stopping is reframed as a *loss* — walking away from a “sure thing” — rather than a natural endpoint. The gap represents £111.49 of additional spend at optimal pricing.

For **skin-seekers**, the sunk cost operates in reverse: having committed to the £213.98 spend from the start, they are anchored to completing all 80 rolls regardless of intermediate outcomes. A skin-seeker who triggers S1 naturally at roll 30 has “wasted” their remaining budget headroom, but a skin-seeker at roll 60 without S1 has already spent £164.22 and faces the classic sunk-cost compulsion to continue.

2. Near-Miss Framing

For **casual players**, with only 16.5% triggering S1 by roll 38, most leave feeling they “almost” got lucky. The variance decomposition confirms this is not paranoia: S1 timing genuinely is the dominant factor separating good outcomes from bad ones in the early-to-mid session (68.9% of explained variance at roll 50). A lower cap would resolve this uncertainty earlier, collapsing the emotional hook before it becomes entangled with spending decisions.

For **skin-seekers**, near-miss framing operates throughout their entire session. Every roll that does not trigger S1 naturally (at 0.5% per roll) reinforces the feeling that the next roll *could* be the one — even though the simulation shows the median player never triggers S1 before the pity cap forces it. The 79-roll uncertainty window means the skin-seeker spends their entire journey in a state of manufactured near-miss anticipation.

3. Discrete Invisibility

Each 10-roll cap increase adds only 11–14 anxious rolls. No single change feels dramatic. A designer moving the cap from 40 to 50 adds 11 rolls of uncertainty — approximately £31.50 in

potential spend — an increment that would be difficult for players to perceive, let alone protest. But the cumulative effect from cap 10 to cap 80 is 79 extra anxious rolls worth £212.72.

This is the core of the discrete goalpost widening effect: a series of individually invisible adjustments that, in aggregate, reshape the entire psychological landscape of a player’s session.

4.4 The Isolation Property

The interaction effects from the 2^3 factorial analysis deserve emphasis. All $S1 \times S2$, $S1 \times S3$, $S2 \times S3$, and $S1 \times S2 \times S3$ interaction terms are below 1.5% of baseline variance at every checkpoint examined. This means that the pity cap operates as an *independent lever*: adjusting it does not create cascading effects on S2 completion dynamics or S3 item selection. From a system design perspective, this is a precision tool. The designer can tune spending pressure with confidence that only the intended variable is affected.

5 What a Fairer System Would Look Like

The analysis also reveals what the system *could* be.

At a pity cap of 30, the uncertainty window spans rolls 18–37 with a width of 20 rolls and a peak S1 variance share of 49.6%. This places the resolution of outcome uncertainty approximately at the 150-point redemption threshold (roll ~ 38): by the time a casual player reaches their exit point, the question of whether they were “lucky” has been substantially answered. For skin-seekers, a cap of 30 means the max prize costs £84.97 rather than £213.98 — the guaranteed outcome is delivered within a spend comparable to what casual players already accept.

At cap 40, the window is 32 rolls wide (rolls 18–49), still resolving within a reasonable horizon. The cost to reach the pity guarantee is £108.24 — comparable to the casual player’s spend at the 150-point threshold. Skin-seekers would pay roughly what casual players already pay, rather than more than double.

Either of these configurations would preserve the consumer-facing function of the pity mechanic (protection against extreme bad luck) while substantially reducing its secondary function as a spending pressure tool. Casual players would still experience uncertainty — this is inherent to any probabilistic system — but that uncertainty would not be *architecturally extended* to coincide with every spending decision they make during their session. Skin-seekers would face a price proportional to the game’s natural economy rather than one inflated by 79 additional anxious rolls.

6 Limitations

Several limitations should be noted.

First, the analysis identifies two player archetypes — casual players who exit at the 150-point redemption threshold and skin-seekers who commit to the pity cap — but real player behaviour is a continuum. Some players stop before 150 points, others fall between the two archetypes, and whale spending patterns differ qualitatively. Modelling a distribution of player types and their interaction with the uncertainty window would strengthen the conclusions.

Second, the psychological mechanisms described — sunk cost, near-miss framing, discrete invisibility — are inferred from the statistical structure rather than measured directly through player data or behavioural experiments. The simulation demonstrates that the *conditions* for these effects exist; it does not prove that players experience them as described.

Third, the model treats each roll as an independent spending decision. In practice, players purchase RP in discrete packages and pre-commit to a batch of rolls. As shown in Section 4.2, this does not weaken the spending pressure argument — the package granularity introduces a second layer of extraction by forcing overspend and creating residual balances. However, the analysis does not model how *package boundary decisions* (“do I buy another package?”) interact with the uncertainty window. The psychological dynamics at these discrete purchase points may differ from the continuous-roll model assumed here.

Fourth, only the S1 pity cap is varied. The S2 pity system (not explored here) introduces additional interactions that may modify the overall spending pressure landscape at higher roll counts.

7 Conclusion

This analysis demonstrates that the S1 pity cap in the Sanctum loot system functions as a monetisation lever with three convergent properties:

1. **Variance dominance.** S1 timing accounts for up to 68.9% of explained outcome variance in the critical early-to-mid session. Whether a player’s experience feels lucky or unlucky is overwhelmingly determined by a single binary event.
2. **Window extension.** Each 10-roll increase in the pity cap extends the zone of peak uncertainty by 11–14 rolls. At cap 80, this zone spans the player’s entire natural session.
3. **Clean isolation.** The pity cap operates independently of S2 and S3 dynamics, with all interaction effects below 1.5% of baseline variance. It is a precision instrument.

The cumulative effect — discrete goalpost widening — transforms a consumer protection mechanic into a spending optimisation tool that operates on both player archetypes simultaneously. For casual players (targeting the 150-point redemption threshold), the cap at 80 rolls creates £111.49 of sunk-cost distance beyond their exit point and ensures that 83.5% leave inside the uncertainty window. For skin-seekers (targeting the max prize), the cap sets the price of the guaranteed outcome at £213.98 rather than the £84.97–£108.24 it would cost at caps of 30–40 — a surcharge applied entirely through the uncertainty window. In both cases, the extraction is achieved through increments that are individually invisible.

The data suggests that a cap of 30–40 rolls would preserve the protective function of the pity system while resolving outcome uncertainty within the casual player’s session and pricing the max prize proportionally to the game’s natural economy. The difference between cap 40 and cap 80 is not player protection — it is £105.74 of additional spending pressure per player, applied through a mechanism that is statistically precise and psychologically difficult to perceive.